Humidity, QNH Altimeter Setting and Dew-Point Prediction using LightGBM Gr

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EXECUTIVE ABSTRACT & SYSTEM SCOPE

Individual thesis investigating atmospheric moisture parameters, barometric QNH pressure tendencies, and dew point prediction using LightGBM. Derives physically consistent relative humidity and assesses atmospheric stability indices.

1. Thermodynamic Coupling of Dew Point and Pressure

Dew point depression (T - Td spread) is the primary physical indicator of impending saturation and cloud formation.

Directly predicting relative humidity often causes physical contradictions; this study predicts dew point and barometric pressure first, then derives relative humidity mathematically.

2. LightGBM Architecture & Feature Dynamics

LightGBM was selected for its leaf-wise tree growth and speed on continuous meteorological time series.

Rolling statistics, pressure gradients, and diurnal solar radiation proxies were engineered to capture thermodynamic shifts.

3. Quantitative Evaluation and Guidance Output

" Dew Point R2: 0.66, MAE: 0.58 deg C

" QNH Altimeter Setting MAE: 1.8 hPa

" Successfully powers automated weather briefing and stability forecast documents for flight planning.